Nametable Mapping

File: nametable_mapping.kicad_sch

CPU Address Bus

File: cpuAddressBus.kicad_sch

Register Access

File: register_access.kicad_sch

PRG Banking

File: prg_banking.kicad_sch

PPU Address Bus

File: ppu_address_bus.kicad_sch

Reset Detection

File: reset_detection.kicad_sch

CPU Data Bus

File: cpu_data_bus.kicad_sch

PCM DAC

File: pcm_dac.kicad_sch

Scanline Detection

File: scanline_detection.kicad_sch

V-Split

File: v_split.kicad_sch

PPU Fetch Counting

File: ppu_fetch_counting.kicad_sch

CHR A0,A1,A2

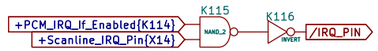
File: chr_a0_a1_a2.kicad_sch

ExRAM Mode

File: exram_mode.kicad_sch

Fill Mode

File: fill_mode.kicad_sch



Pulse Length Lookup Table

File: pulse_length_lookup_table.kicad_sch

Pulse Timer

File: pulse_timer.kicad_sch

Pulse Duty Envelope Volume

File: pulse_duty_envelope_volume.kicad_sch

Pulse DAC

File: Pulse DAC.kicad_sch

PPU Data Bus

File: ppu_data_bus.kicad_sch

CHR Bank Registers

File: chr_bank_registers.kicad_sch

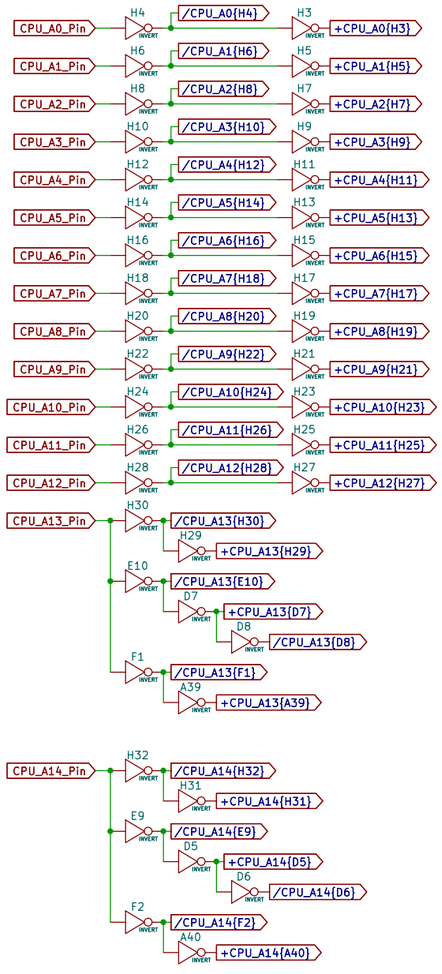
CHR Bank Selection Logic

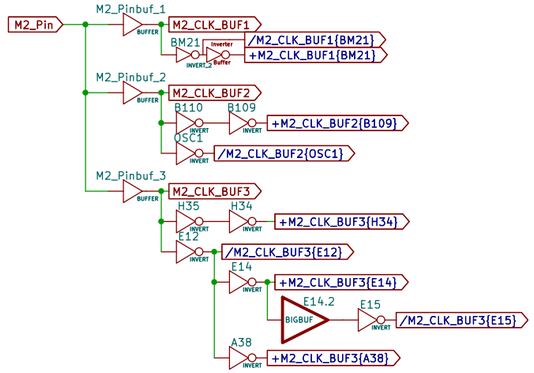
File: chr_bank_selection_logic.kicad_sch

CHR Address Bus

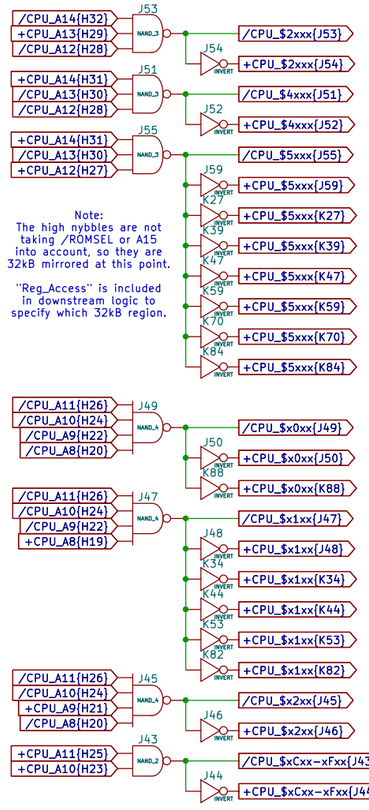
File: chr_address_bus.kicad_sch

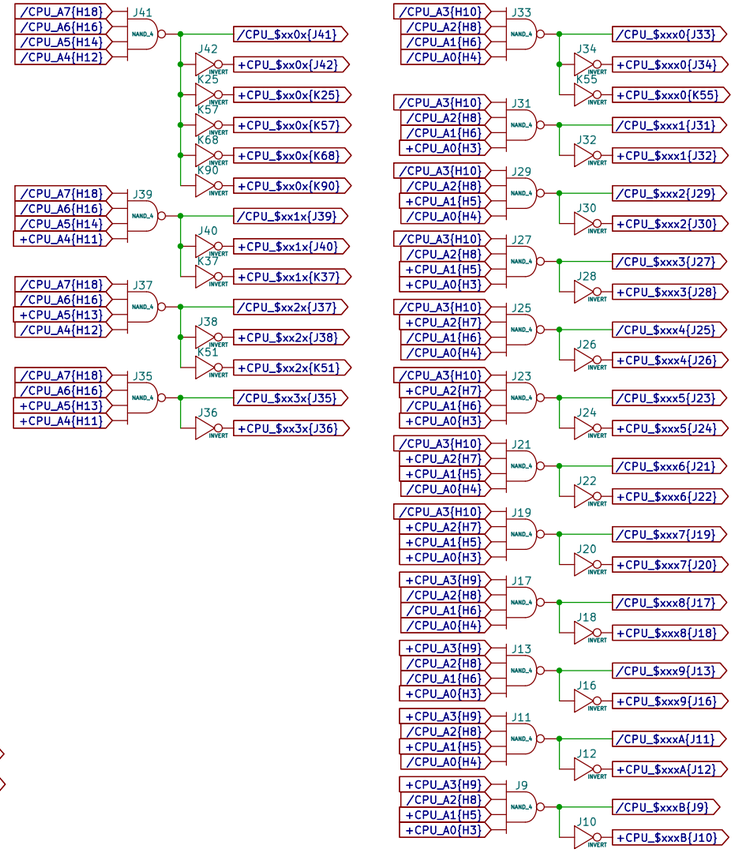
CPU Address Bus Buffering

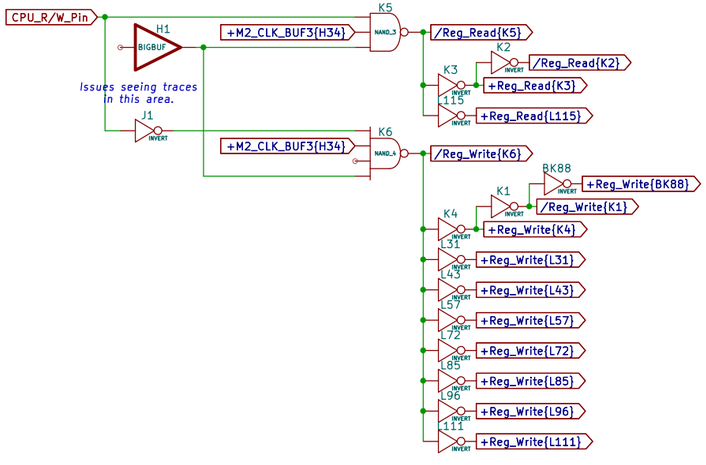


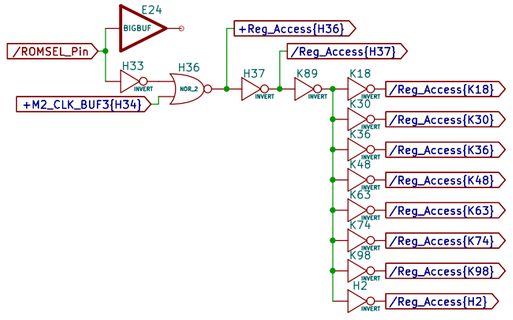


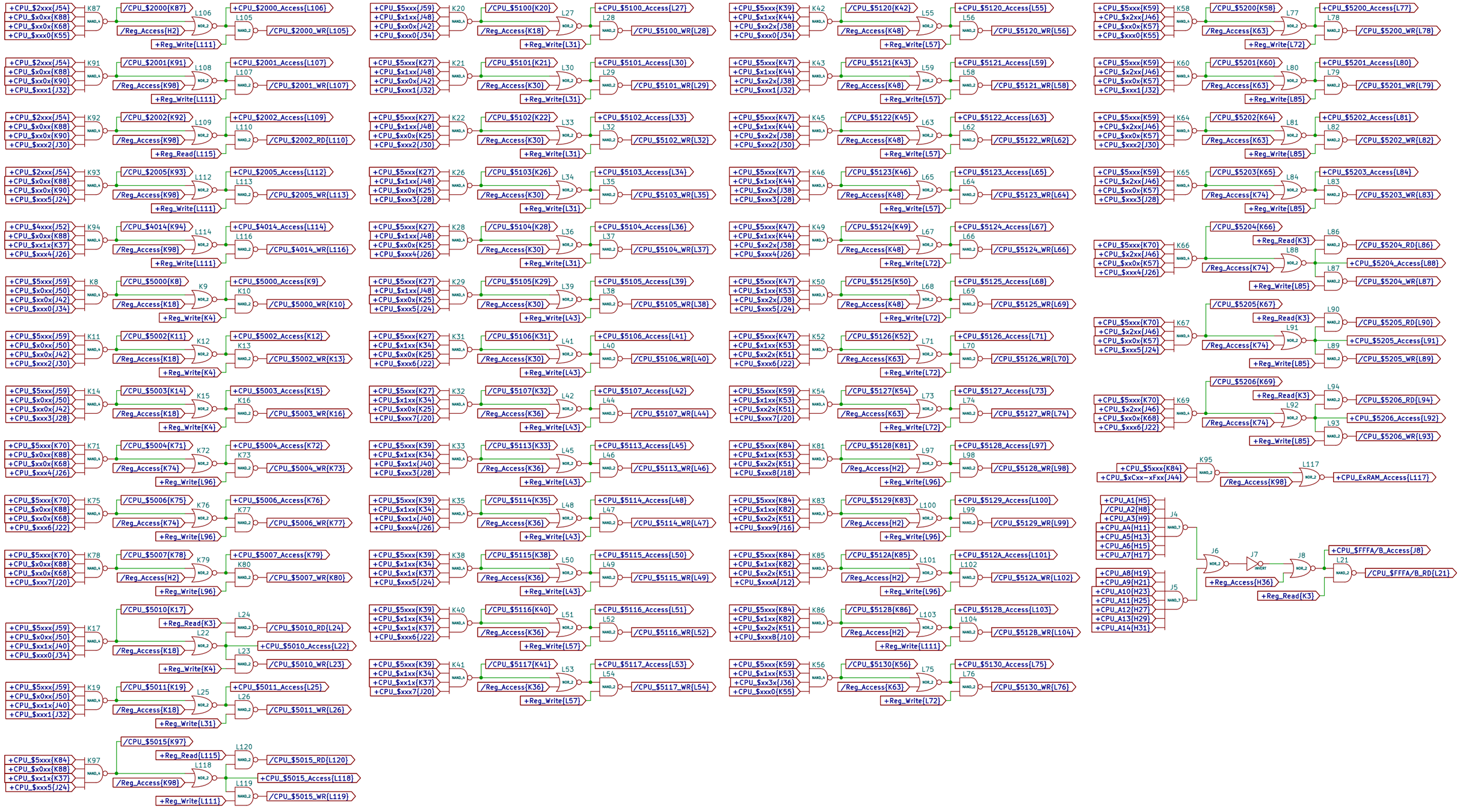
CPU Address Bus Nybble Decoding







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L105
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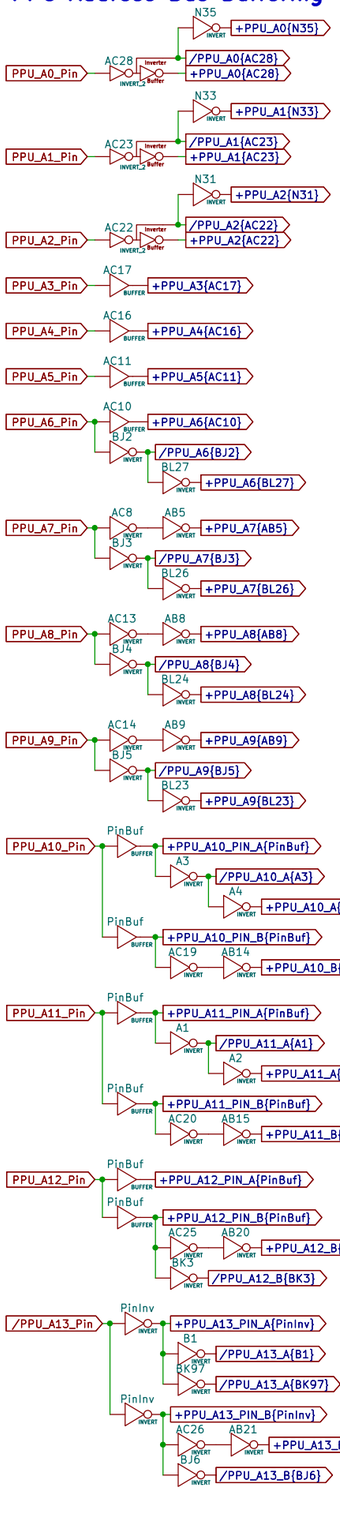
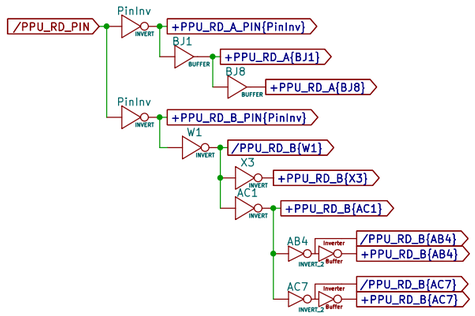
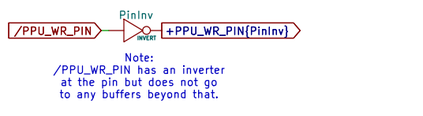
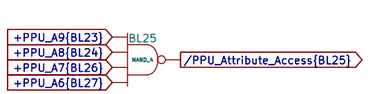
+CPU_$5206_Access{L92}
/CPU_$5206_RD{L94}
/CPU_$5206_WR{L93}
+CPU_$5xxx{K84}
+CPU_$xCxx-xFxx{J44}
K95
NAND_2
/Reg_Access{K98}
L117
NOR_2
+CPU_ExRAM_Access{L117}
+CPU_A1{H5}
/CPU_A2{H8}
+CPU_A3{H9}
+CPU_A4{H11}
+CPU_A5{H13}
+CPU_A6{H15}
+CPU_A7{H17}
J4
NAND_7
+CPU_A8{H19}
+CPU_A9{H21}
+CPU_A10{H23}
+CPU_A11{H25}
+CPU_A12{H27}
+CPU_A13{H29}
+CPU_A14{H31}
J5
NAND_7
J6
NOR_2
J7
INVERT
J8
NOR_2
+Reg_Access{H36}
+Reg_Read{K3}
+CPU_$FFFA/B_Access{J8}
L21
NAND_2
/CPU_$FFFA/B_RD{L21}

PPU Address Bus Buffering

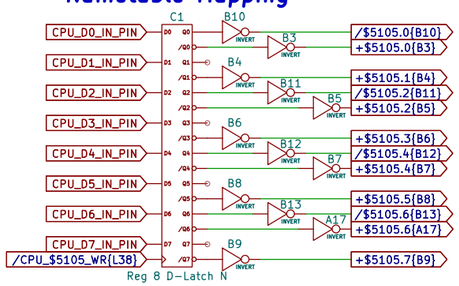




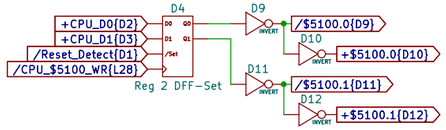




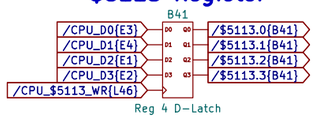
$5105 Register Nametable Mapping



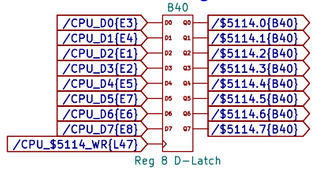
$5100 Register
PRG Mode



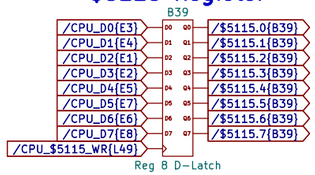
PRG Bank Registers
$5113 Register



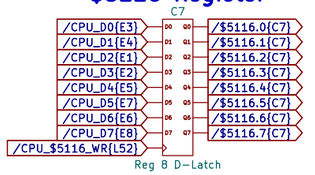
$5114 Register



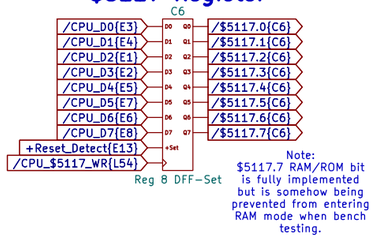
$5115 Register



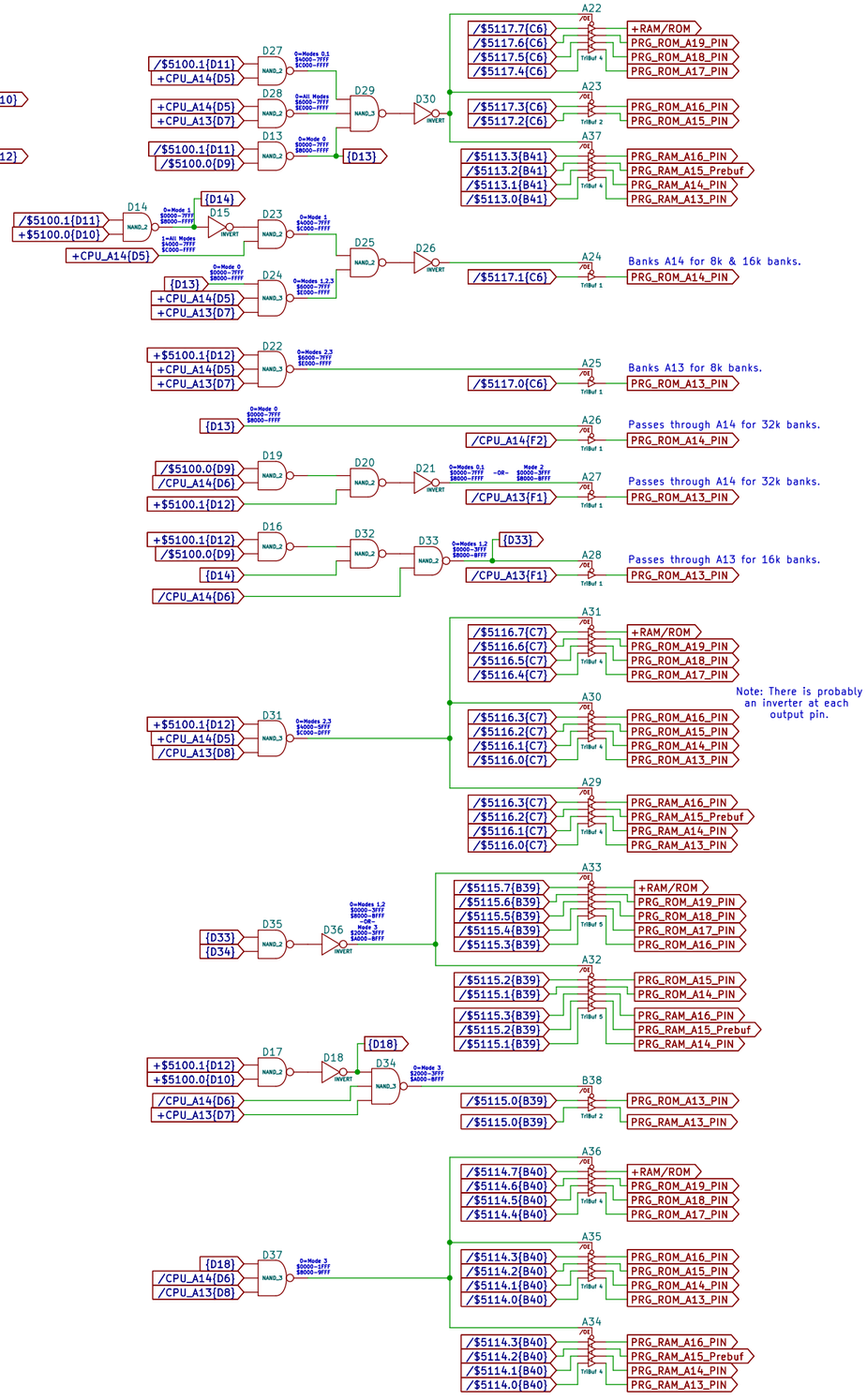
$5116 Register

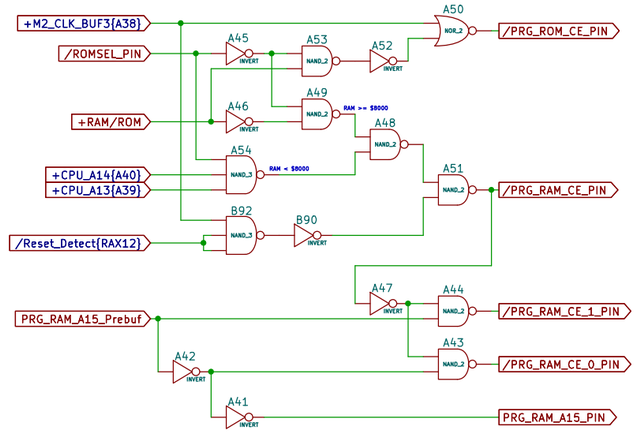


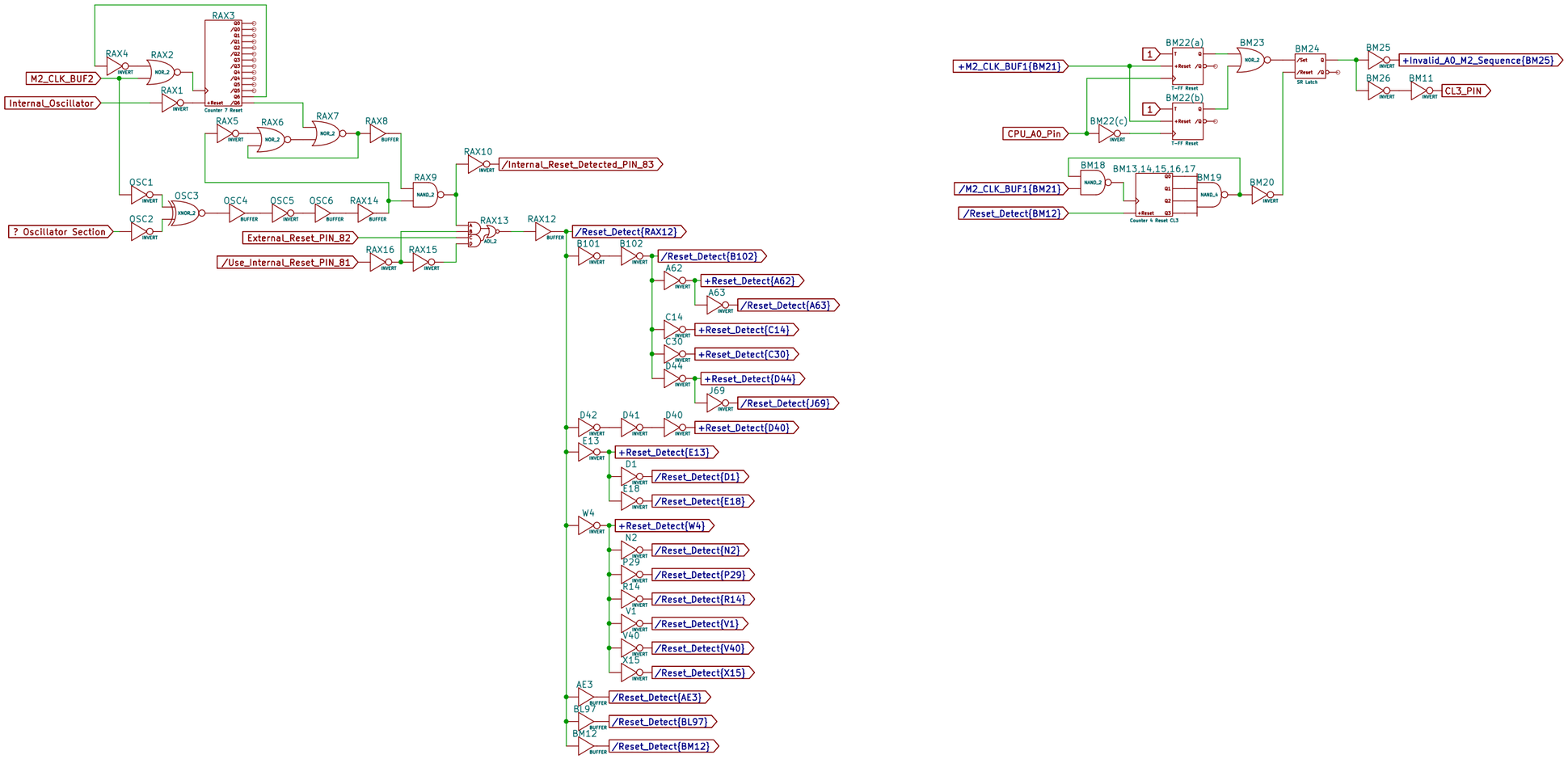
$5117 Register



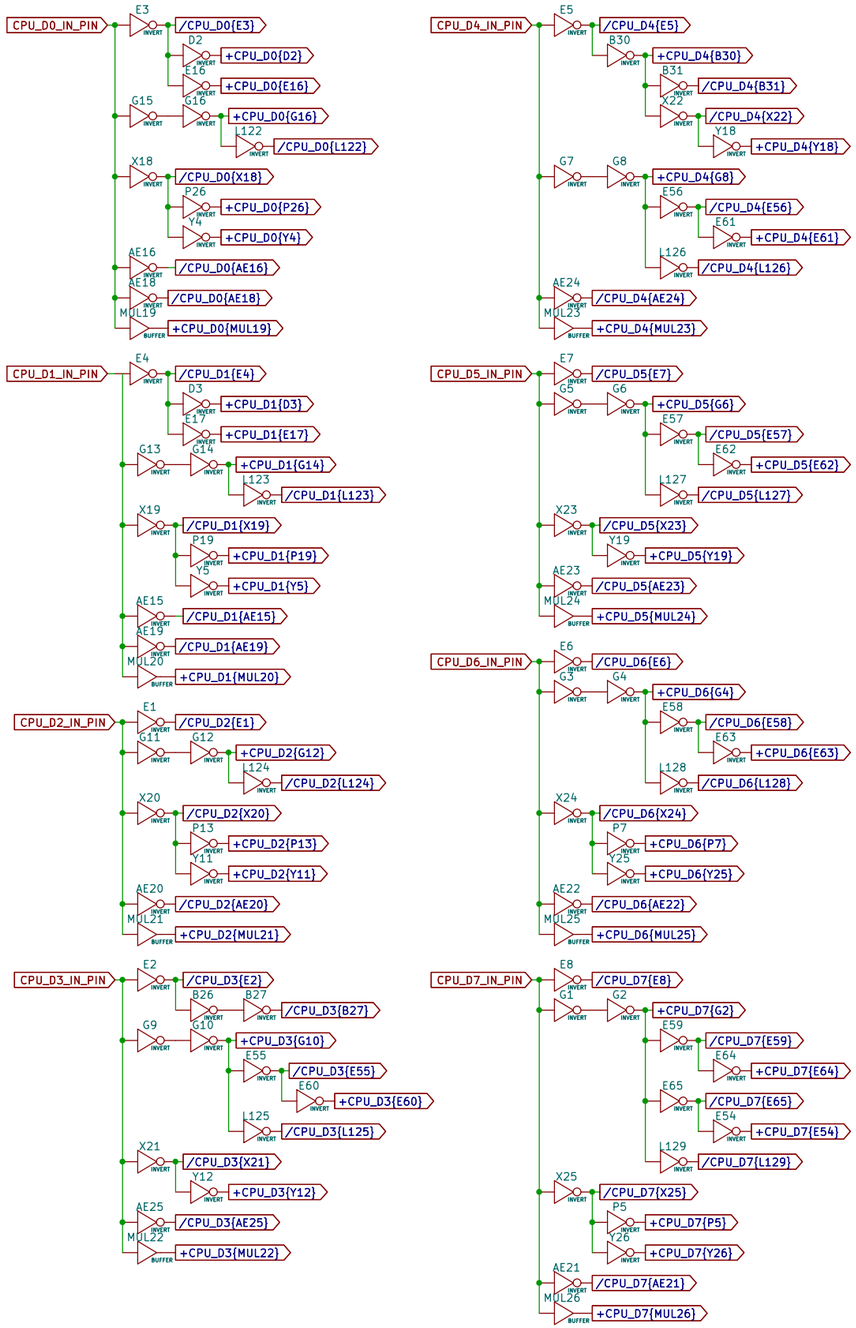
Note: $5117.7 RAM/ROM bit is fully implemented but is somehow being prevented from entering RAM mode when bench testing.



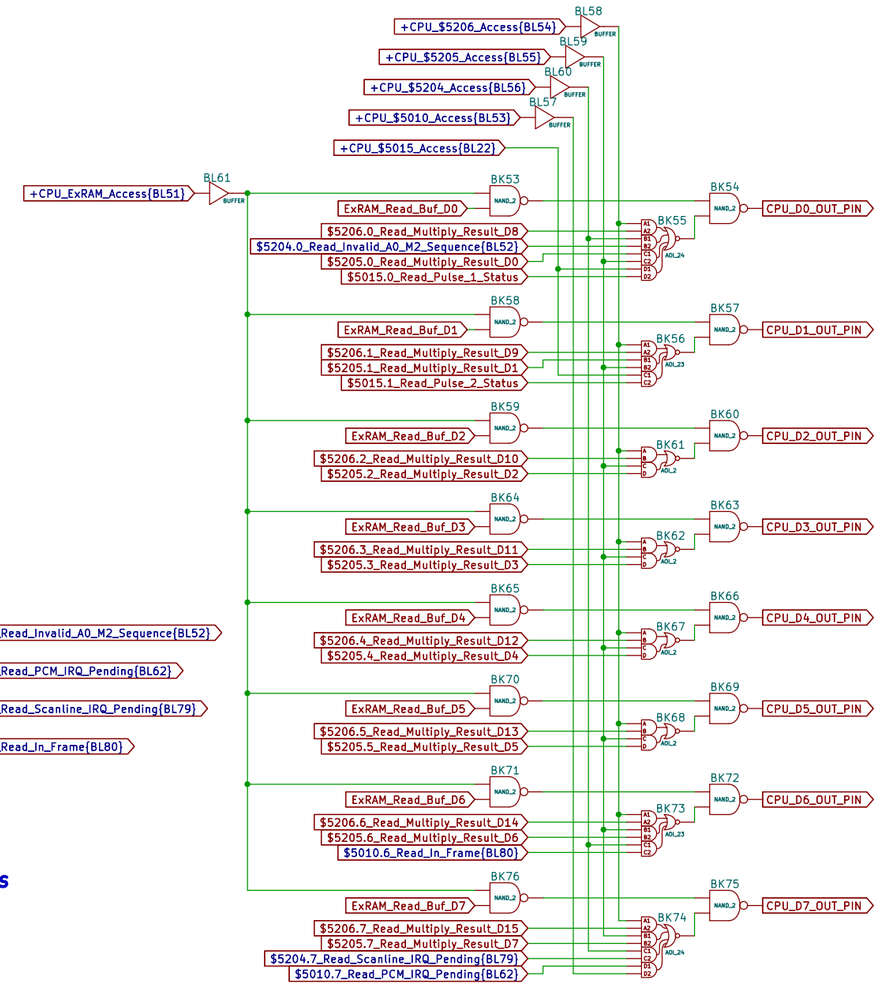



RAX3
RAX4
RAX2
RAX1
M2_CLK_BUF2
Internal_Oscillator
Counter 7 Reset
RAX5
RAX6
RAX7
RAX8
RAX10
/Internal_Reset_Detected_PIN_83
RAX9
OSC1
OSC3
OSC4
OSC5
OSC6
RAX14
OSC2
? Oscillator Section
External_Reset_PIN_82
/Use_Internal_Reset_PIN_81
RAX16
RAX15
RAX13
RAX12
/Reset_Detect{RAX12}
B101
B102
/Reset_Detect{B102}
A62
+Reset_Detect{A62}
A63
/Reset_Detect{A63}
C14
+Reset_Detect{C14}
C30
+Reset_Detect{C30}
D44
+Reset_Detect{D44}
J69
/Reset_Detect{J69}
D42
D41
D40
+Reset_Detect{D40}
E13
+Reset_Detect{E13}
D1
/Reset_Detect{D1}
E18
/Reset_Detect{E18}
W4
+Reset_Detect{W4}
N2
/Reset_Detect{N2}
P29
/Reset_Detect{P29}
R14
/Reset_Detect{R14}
V1
/Reset_Detect{V1}
V40
/Reset_Detect{V40}
X15
/Reset_Detect{X15}
AE3
/Reset_Detect{AE3}
BL97
/Reset_Detect{BL97}
BM12
/Reset_Detect{BM12}
+M2_CLK_BUF1{BM21}
BM22(a)
BM23
BM24
BM25
+Invalid_A0_M2_Sequence{BM25}
BM26
BM11
CL3_PIN
CPU_A0_Pin
BM22(c)
BM22(b)
T-FF Reset
SR Latch
BM18
BM13,14,15,16,17
BM19
BM20
/M2_CLK_BUF1{BM21}
/Reset_Detect{BM12}
Counter 4 Reset CL3

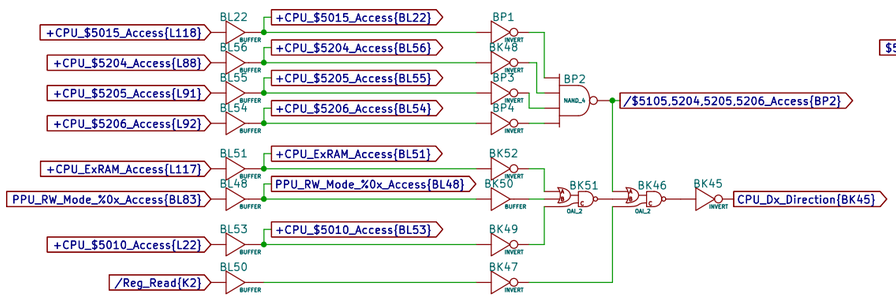
Incoming CPU Data Bus Buffering Bonanza



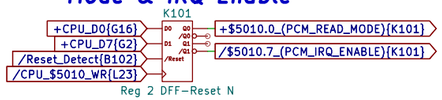
Multiplexing Data to be Driven onto the CPU Data Bus

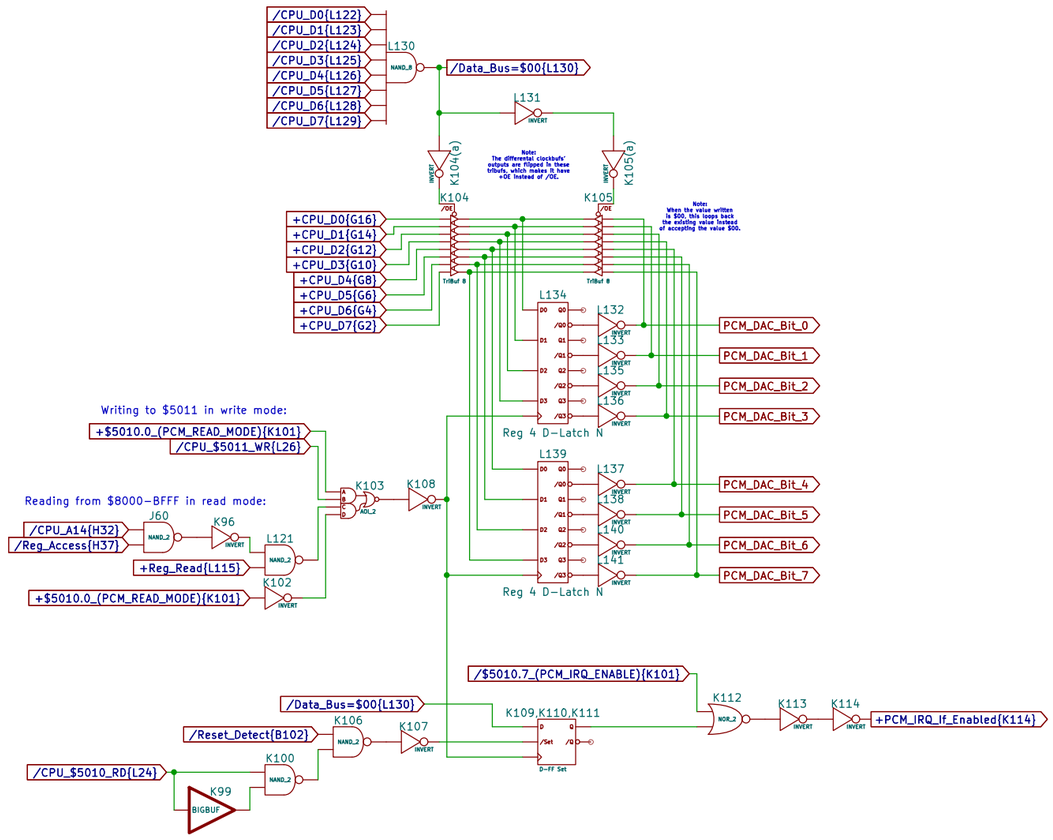


Decides When to Drive the CPU Data Bus



$5010 Register
Mode & IRQ Enable





+PPU_A13_B{AB21}
+PPU_A12_B{AB20}
+PPU_A11_B{AB15}
+PPU_A10_B{AB14}
+PPU_A9{AB9}
+PPU_A8{AB8}
+PPU_A7{AB5}
+PPU_A6{AC10}
+PPU_A5{AC11}
+PPU_A4{AC16}
+PPU_A3{AC17}
+PPU_A2{AC22}
+PPU_A1{AC23}
+PPU_A0{AC28}
+PPU_RD_B{AB4}
/Exit_In_Frame{AC6}
+PPU_RD_B{AC7}

When not in frame, the previous PPU read address always looks like $0000. Since A13, A12 are looking at range $2xxx instead of matches, $0xxx won't match $2xxx. Therefore, scanlines don't get detected when using PPUDATA to read from $0000 during v-blank.

+Use_Internal_Scanline_Detection_PIN_92
External_Scanline_Input_PIN_29
/Use_Internal_Scanline_Detection{AA1}
+Use_Internal_Scanline_Detection{AA1}
/New_Scanline_Detected{AA3}
/New_Scanline_Detected{R2}

High when the present PPU address is from $2xxx and matches the previous read.

Low when the 3 most recent PPU reads were from $2xxx and all matched.

M2_CLK_BUF1
+PPU_RD_B{AC1}
/Exit_In_Frame{AC6}
/PPU_RD_B{W1}

$5204 Register Scanline IRQ Enable

+CPU_D7{Y26}
/Reset_Detect{X15}
/CPU_$5204_WR{L87}
+Scanline_IRQ_Pin{X14}
/CPU_$5204_RD{L86}
+New_Scanline_Detected{AA3 via Z23}
/In_Frame{Z23}
/New_Scanline_Detected{AA3}
/IRQ
/Scanline_IRQ_Status{X28}

$5203 Register IRQ Scanline Compare Value

Y2,7,9,14,16,21,23,28
+CPU_D0{Y4}
+CPU_D1{Y5}
+CPU_D2{Y11}
+CPU_D3{Y12}
+CPU_D4{Y18}
+CPU_D5{Y19}
+CPU_D6{Y25}
+CPU_D7{Y26}
/CPU_$5203_WR{L83}
Reg 8 D-Latch

Hi when $5203 matches the present V-Split scanline.

D'Morgan tells us that we can pretend Y8,Y15,Y22 are all AND gates.

IRQ Scanline Counter

Z2-Z18
/New_Scanline_Detected{AA3}
+In_Frame{Z21}
/In_Frame{Z23}
/Scanline_240_and_up{Z22}

$2002 Register Read Snooping

Latches $2002.7 value read by CPU for 1 CPU cycle.

/CPU_$FFFA/B_RD{L21}
/CPU_$4014_WR{L116}
/Exit_In_Frame{AC6}
/CPU_$2002_RD{L110}
+CPU_D7{Y26}
M2_CLK_BUF1
/New_Scanline_Detected{AA3}
/In_Frame{Z23}
+In_Frame{Z21}

$2001 Register Write Snooping

+CPU_D3{Y12}
+CPU_D4{Y18}
/CPU_$2001_WR{L107}
Hi = PPU Rendering Enabled
+Rendering_In_Frame{Z25}
/Rendering_In_Frame{V39}
/Rendering_In_Frame{R24}
+Rendering_In_Frame{N9}

V−Split Scanline Counter

$5201 Register V−Split V−Scroll

W3,V24,V25,V41

/CPU_D0{X18}
/CPU_D1{X19}
/CPU_D2{X20}
/CPU_D3{X21}
/CPU_D4{X22}
/CPU_D5{X23}
/CPU_D6{X24}
/CPU_D7{X25}
/Reset_Detect{V40}
/CPU_$5201_WR{L79}

Reg 8 DFF−Reset

V−Split_Scanline_Detected{P3}
/Rendering_In_Frame{V39}
Reload_V−Split_Scanline_Count{T1}

V15 V14 V13 T7 T8 T9 +V−Split_Scanline_Counter.0{T9}
V18 V17 V16 T10 T12 T13 +V−Split_Scanline_Counter.1{T13}
V21 V20 V19 T14 T15 T16 +V−Split_Scanline_Counter.2{T16}
V26 V23 V22 T17 T18 T19 +V−Split_Scanline_Counter.3{T19}
V29 V28 V27 T20 T21 T22 +V−Split_Scanline_Counter.4{T22}
V32 V31 V30 T23 T24 T25 +V−Split_Scanline_Counter.5{T25}
V35 V34 V33 T26 T27 T28 +V−Split_Scanline_Counter.6{T28}
V38 V37 V36 T29 T30 T31 +V−Split_Scanline_Counter.7{T31}

T−FF Preload

$5200 Register V−Split Mode & H−Threshold

W2

/CPU_D0{X18}
/CPU_D1{X19}
/CPU_D2{X20}
/CPU_D3{X21}
/CPU_D4{X22}
/Reset_Detect{V1}

/CPU_$5200_WR{L78}
Reg 5 D−Latch−Reset

Note: Bit 5 does not exist.

+CPU_D6{P7}
+CPU_D7{P5}
/Reset_Detect{N2}
/CPU_$5200_WR{L78}
Reg 2 DFF−Reset N

$5200.6{P6}
$5200.7{P6}

+PPU_Fetch_Counter.2{S18}
+PPU_Fetch_Counter.3{S21}
+PPU_Fetch_Counter.4{S24}
+PPU_Fetch_Counter.5{S25}
+PPU_Fetch_Counter.6{S17}
+PPU_Fetch_Counter.7{S29}

V3 V4 V5 V6 V10 V11 V12 V9 V8 V7 T6
V−Split_Side_Being_Rendered{V7}
/Any_PPU_Fetch_While_In_Frame{X1}

D−FF No Reset

+In_Frame{Z21}
+PPU_RD_B{X3}
X2 X1 P2 P1 S5 S6
/Any_PPU_Fetch_While_In_Frame{X1}
R5.Q
Increment_PPU_Fetch_Count{S6}

I am guessing this is any PPU fetch excluding sprites

Increment_PPU_Fetch_Count{S6}
R4.Q1
R6 R7 R8
V−Split_Scanline_Detected_Internal{R8}

/Use_Internal_Scanline_Detection{AA1}
V−Split_Scanline_Detected_Internal{R8}
+Use_Internal_Scanline_Detection{AA1}
V−Split_Scanline_PIN_30
R1 P3 T5
V−Split_Scanline_Detected{P3}
V−Split_Scanline_Detected{T5}

+PPU_Fetch_Count.0{S10}
+PPU_Fetch_Count.1{S13}
+PPU_Fetch_Count.2{S18}
+PPU_Fetch_Count.4{S23}
+PPU_Fetch_Count.7{S29}
S3

+PPU_Fetch_Count.0{S10}
+PPU_Fetch_Count.1{S13}
+PPU_Fetch_Count.2{S18}
+PPU_Fetch_Count.5{S25}
+PPU_Fetch_Count.7{S29}
S4

R4
R4.Q1
/Reset_Detect{R14}
Increment_PPU_Fetch_Count{S6}
Reg 2 DFF−Set

Unknown 3−Bit Scanline Counter

$2005 Register 2nd Write Snooping

P12,13,18,19,25,26

/CPU_D0{X18}
/CPU_D1{X19}
/CPU_D2{X20}
/Reset_Detect{P29}
Reg 3 DFF−Reset

/CPU_$2005_WR{L113}
R29 R26 P28
/CPU_$5130_WR{L76}
R28 R27
D−FF Reset

Resyncs 2nd write tracking when $5130 is written.

/Q −> D turns this into a T−FF with async /reset.

V−Split_Scanline_Detected{P3}
/Rendering_In_Frame{R24}

P22 P24 P23 P21 P20 +3−Bit_Counter.0{P20}
P17 P16 P15 P14 +3−Bit_Counter.1{P14}
P11 P10 P9 P8 +3−Bit_Counter.2{P8}

T−FF Preload

+V−Split_Scanline_Counter.0{T9}
+V−Split_Scanline_Counter.1{T13}
+V−Split_Scanline_Counter.2{T16}
+V−Split_Scanline_Counter.3{T19}
+V−Split_Scanline_Counter.5{T25}
+V−Split_Scanline_Counter.6{T28}
+V−Split_Scanline_Counter.7{T31}
S30

/V−Split_Scanline=239{S30}
+Rendering_In_Frame{Z25}
/Reset_Detect{V1}
BH29 BH28 BH30
This cell has unknown logic

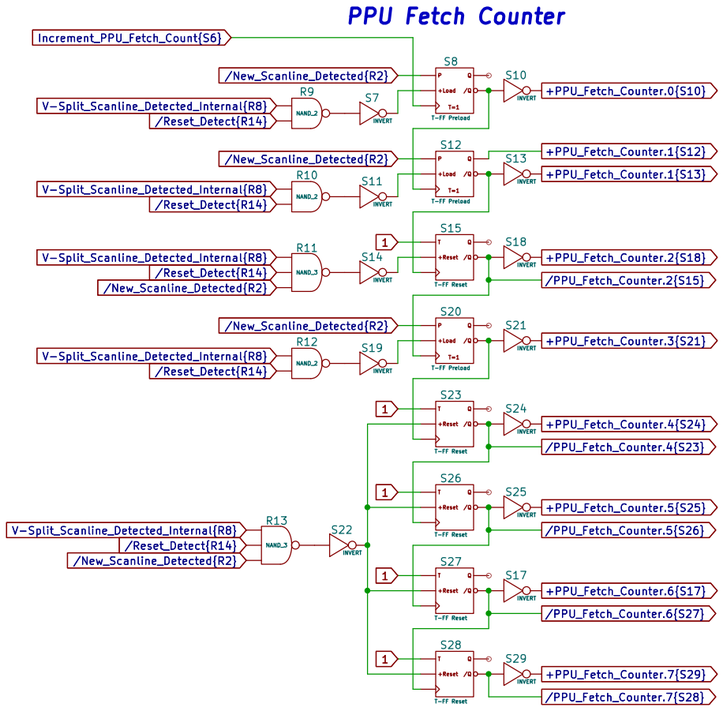
+Rendering_In_Frame{Z25}
V−Split_Scanline_Detected{T5}
BH31 BH32

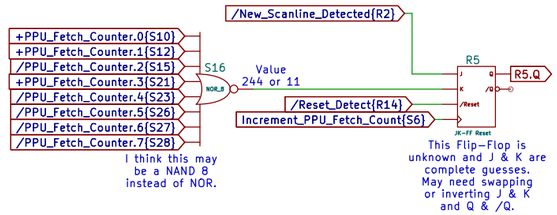
V−Split_Scanline_Detected{T5}
T3 T2 T1
Reload_V−Split_Scanline_Count{T1}
T4

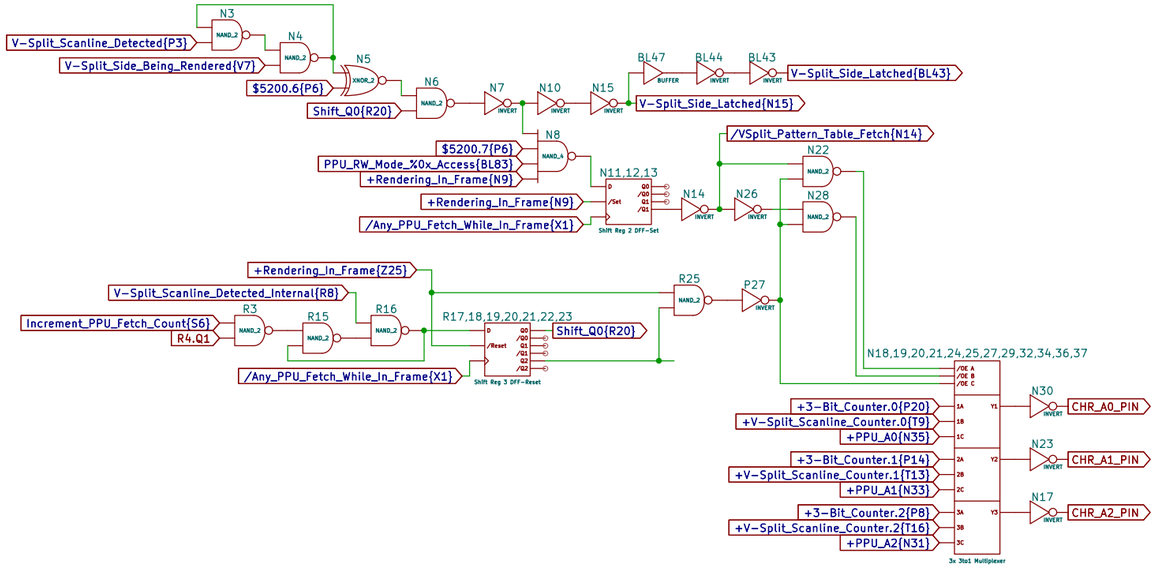
JK−FF Reset

This Flip−Flop is unknown and J & K are complete guesses. May need swapping or inverting J & K and Q & /Q.

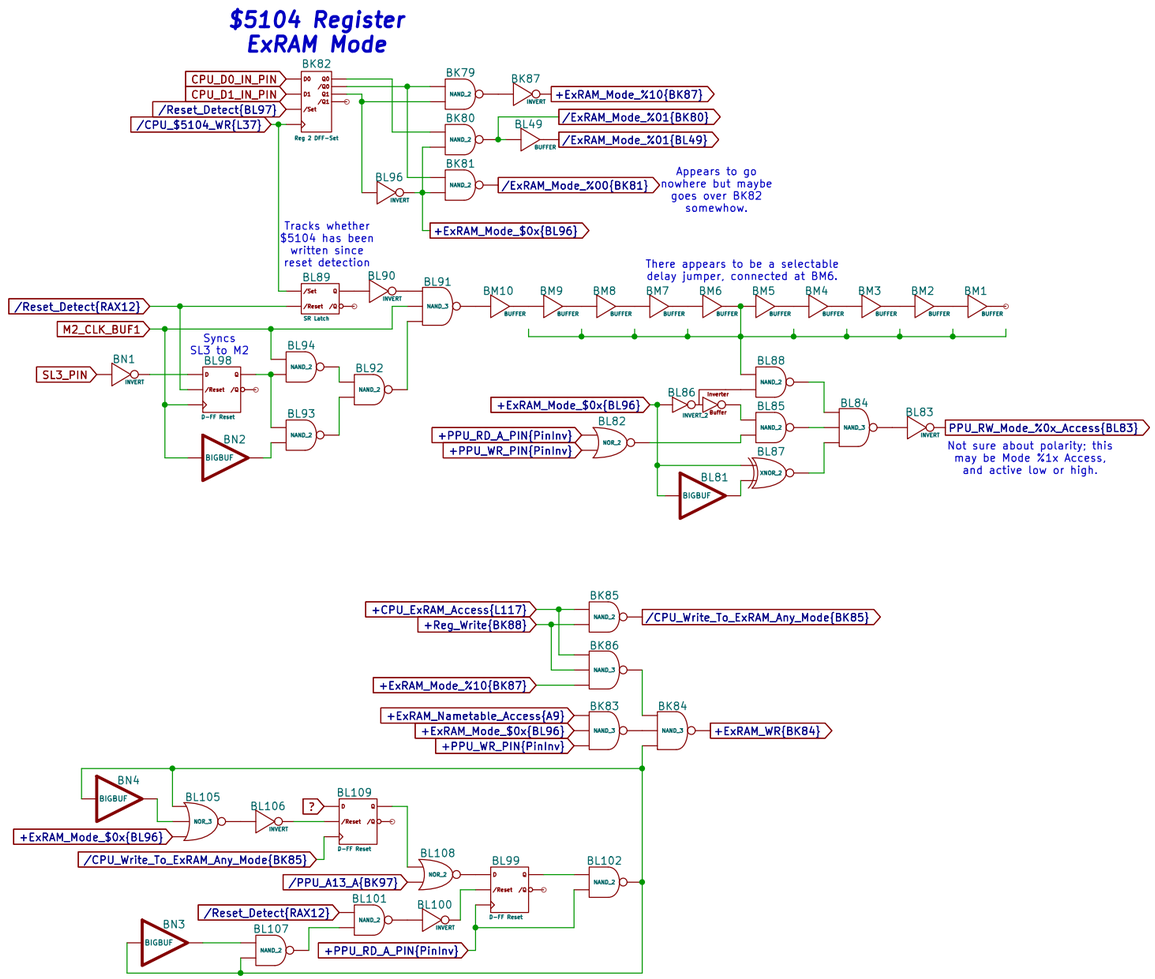
PPU Fetch Counter

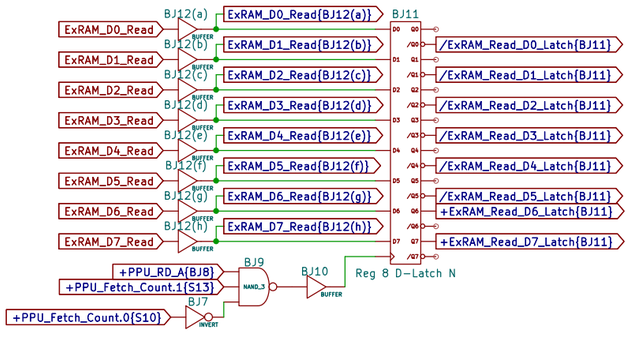




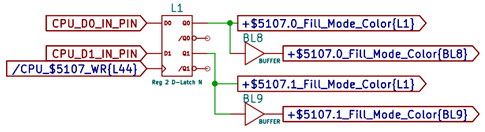

N3
NAND_2
V-Split_Scanline_Detected{P3}
N4
NAND_2
V-Split_Side_Being_Rendered{V7}
N5
XNOR_2
$5200.6{P6}
N6
NAND_2
Shift_Q0{R20}
N7
INVERT
N10
INVERT
N15
INVERT
BL47
BUFFER
BL44
INVERT
BL43
INVERT
V-Split_Side_Latched{BL43}
V-Split_Side_Latched{N15}
N8
NAND_4
$5200.7{P6}
PPU_RW_Mode_%0x_Access{BL83}
+Rendering_In_Frame{N9}
N11,12,13
D
/Set
Q0
/Q0
Q1
/Q1
Shift Reg 2 DFF-Set
+Rendering_In_Frame{N9}
/Any_PPU_Fetch_While_In_Frame{X1}
N14
INVERT
N26
INVERT
/VSplit_Pattern_Table_Fetch{N14}
N22
NAND_2
N28
NAND_2
+Rendering_In_Frame{Z25}
V-Split_Scanline_Detected_Internal{R8}
Increment_PPU_Fetch_Count{S6}
R4.Q1
R3
NAND_2
R15
NAND_2
R16
NAND_2
R17,18,19,20,21,22,23
D
/Reset
Q0
/Q0
Q1
/Q1
Q2
/Q2
Shift Reg 3 DFF-Reset
Shift_Q0{R20}
/Any_PPU_Fetch_While_In_Frame{X1}
R25
NAND_2
P27
INVERT
N18,19,20,21,24,25,27,29,32,34,36,37
/OE A
/OE B
/OE C
+3-Bit_Counter.0{P20}
+V-Split_Scanline_Counter.0{T9}
+PPU_A0{N35}
1A
1B
1C
Y1
N30
INVERT
CHR_A0_PIN
+3-Bit_Counter.1{P14}
+V-Split_Scanline_Counter.1{T13}
+PPU_A1{N33}
2A
2B
2C
Y2
N23
INVERT
CHR_A1_PIN
+3-Bit_Counter.2{P8}
+V-Split_Scanline_Counter.2{T16}
+PPU_A2{N31}
3A
3B
3C
Y3
N17
INVERT
CHR_A2_PIN
3x 3to1 Multiplexer

$5104 Register
ExRAM Mode

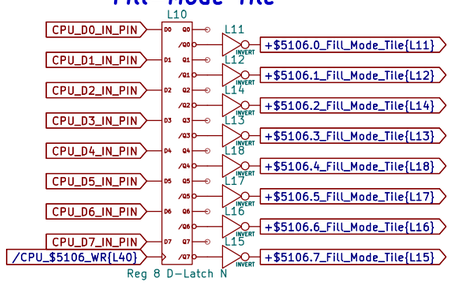




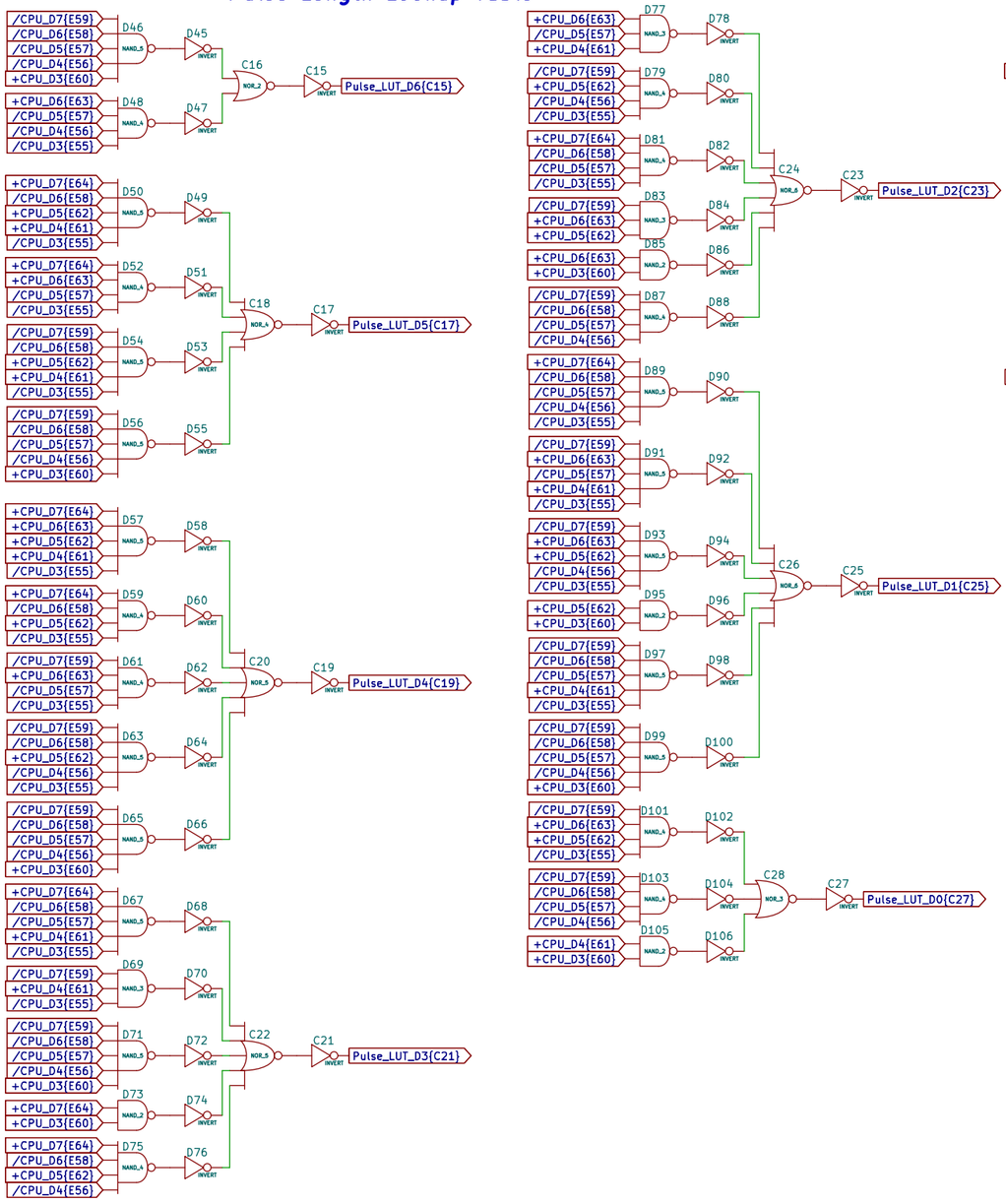
$5107 Register
Fill-Mode Color



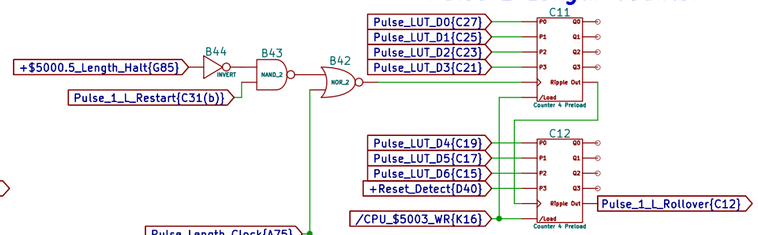
$5106 Register
Fill-Mode Tile



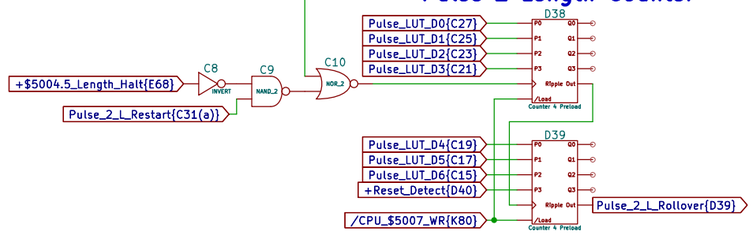
5-Bit to 7-Bit Pulse Length Lookup Table



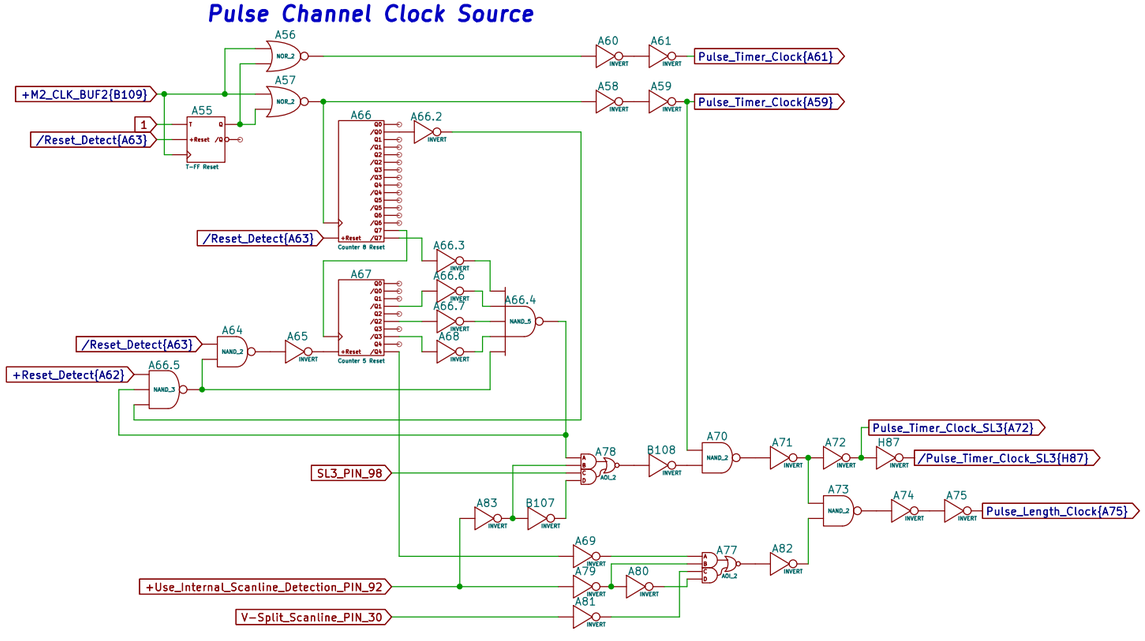
$5003 Register Pulse 1 Length Counter



$5007 Register Pulse 2 Length Counter



Pulse Channel Clock Source



$5002 Register
Pulse 1 Timer

Pulse 1
T Counter

G83
+CPU_D0{G16}
+CPU_D1{G14}
+CPU_D2{G12}
+CPU_D3{G10}
Reg 4 D-Latch N

G84
+CPU_D4{G8}
+CPU_D5{G6}
+CPU_D6{G4}
+CPU_D7{G2}
/CPU_$5002_WR{K13}
Reg 4 D-Latch N

J70
J66
H109
Pulse_Timer_Clock{A59}
Counter 4 Preload

H93
NAND_3
0
Pulse_1_T_Rollover{H109}

$5003 Register

E67
+CPU_D0{G16}
+CPU_D1{G14}
+CPU_D2{G12}
/CPU_$5003_WR{K16}
Reg 3 D-Latch N

H82(b)
/Reset_Detect{B102}
Pulse_Timer_Clock{A61}
D-FF Reset
/Pulse_1_T_Restart{H82(b)}

/Pulse_Timer_Clock_SL3{H87}
H97
NAND_2

H88
INVERT
/CPU_$5003_WR{K16}
H83
NOR_2
H83.2
Unk 2to2
Pulse_Timer_Clock_SL3{A72}

Pulse 1
Envelope Counter

/$5000.5_Length_Halt{G85}
G91
NAND_2
G100
Pulse_1_Envelope.0{G100}
Pulse_1_Envelope.1{G100}
Pulse_1_Envelope.2{G100}
Pulse_1_Envelope.3{G100}
Pulse_1_Envelope_Rollover{G100}
Counter 4 Preload

H85
H86
H84
H78
INVERT
NOR_2
Pulse_1_Envelope_Rollover{G100}
Pulse_Timer_Clock{A61}

H82(a)
Reg 3 D-Latch N

H94
NOR_2
H95
INVERT
H96
H98
G92

+Q0{H80 via H52}
+Q1{H80 via H60}
+Q2{H80 via H68}
+Q3{H80 via H76}
H77
NAND_4

$5006 Register
Pulse 2 Timer

Pulse 2
T Counter

E40
+CPU_D0{G16}
+CPU_D1{G14}
+CPU_D2{G12}
+CPU_D3{G10}
Reg 4 D-Latch N

E41
+CPU_D4{G8}
+CPU_D5{G6}
+CPU_D6{G4}
+CPU_D7{G2}
/CPU_$5006_WR{K77}
Reg 4 D-Latch N

F4
F3
F6
Pulse_Timer_Clock{A59}
Counter 4 Preload

E44
NAND_3
0
Pulse_2_T_Rollover{F6}

$5007 Register

G23
+CPU_D0{G16}
+CPU_D1{G14}
+CPU_D2{G12}
/CPU_$5007_WR{K80}
Reg 3 D-Latch N

E34(a)
/Reset_Detect{B102}
Pulse_Timer_Clock{A61}
D-FF Reset
/Pulse_2_T_Restart{E34(a)}

/Pulse_Timer_Clock_SL3{H87}
E52
NAND_2

F5
INVERT
/CPU_$5007_WR{K80}
E38
NOR_2
E37
Unk 2to2
Pulse_Timer_Clock_SL3{A72}

Pulse 2
Envelope Counter

/$5004.5_Length_Halt{E68}
G21
NAND_2
F7
Pulse_2_Envelope.0{F7}
Pulse_2_Envelope.1{F7}
Pulse_2_Envelope.2{F7}
Pulse_2_Envelope.3{F7}
Pulse_2_Envelope_Rollover{F7}
Counter 4 Preload

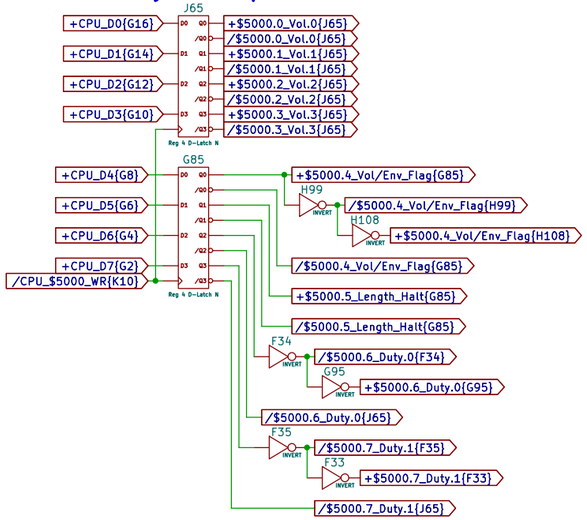
E36
E49
E34(b)
E43
E50
E42
INVERT
NOR_2
Pulse_2_Envelope_Rollover{F7}
Pulse_Timer_Clock{A61}

E35
E39
G22

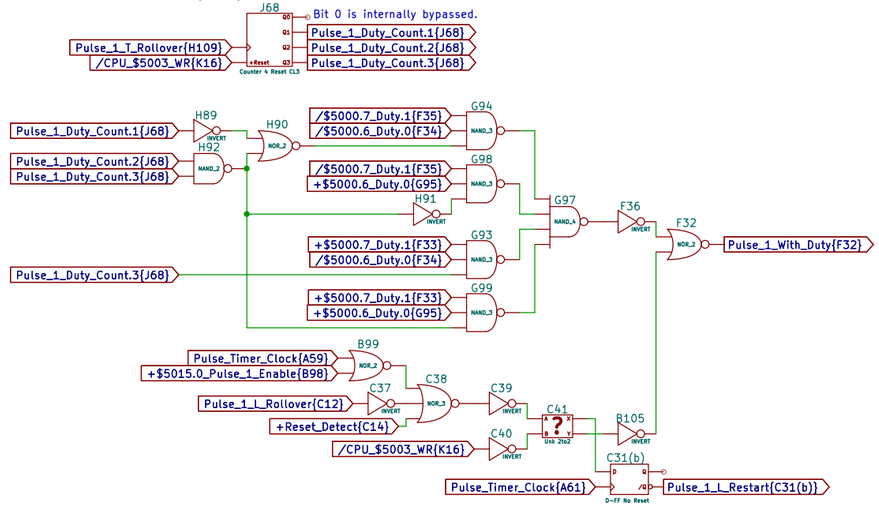
+Q0{H80 via H52}
+Q1{H80 via H60}
+Q2{H80 via H68}
+Q3{H80 via H76}
G43
NAND_4

G44
Reg 3 D-Latch N

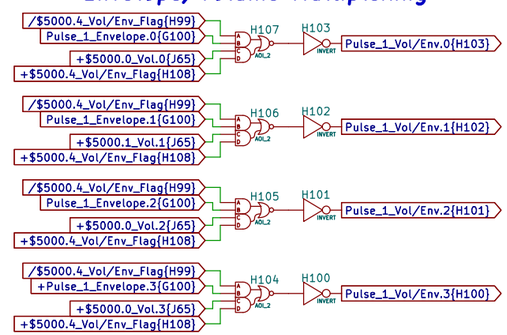
$5000 Register
Pulse 1 Duty Envelope Volume



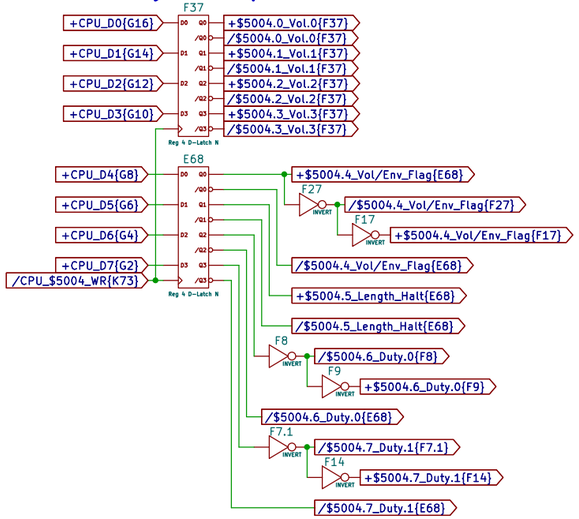
Pulse 1
Duty Cycle Counter



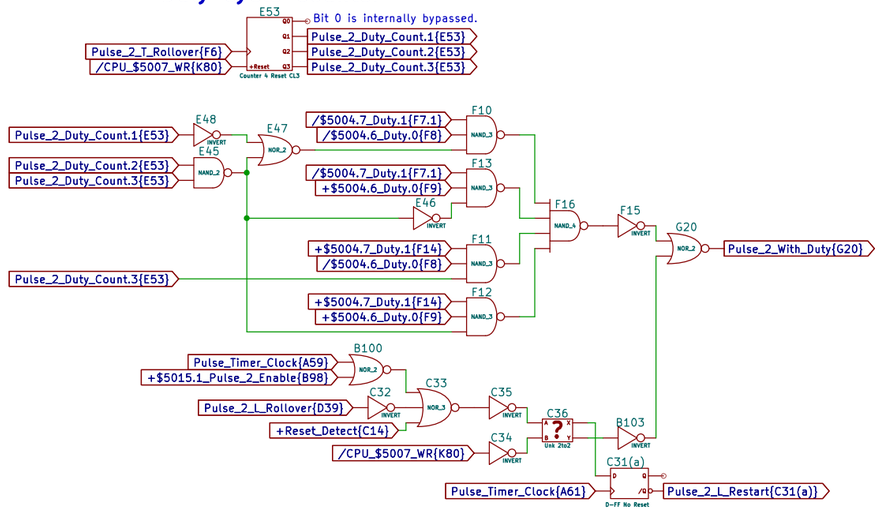
Pulse 1
Envelope/Volume Multiplexing



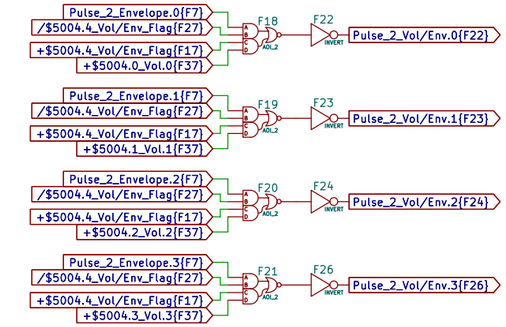
$5004 Register
Pulse 2 Duty Envelope Volume



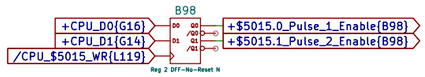
Pulse 2
Duty Cycle Counter



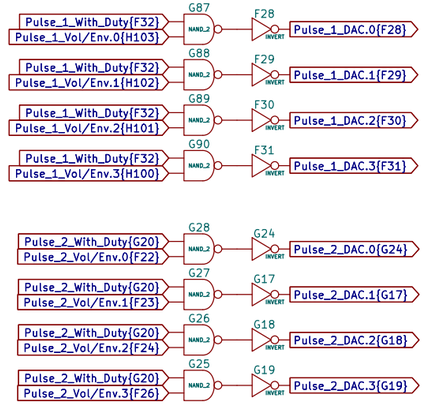
Pulse 2
Envelope/Volume Multiplexing



$5015 Write Register
Pulse 1&2 Status

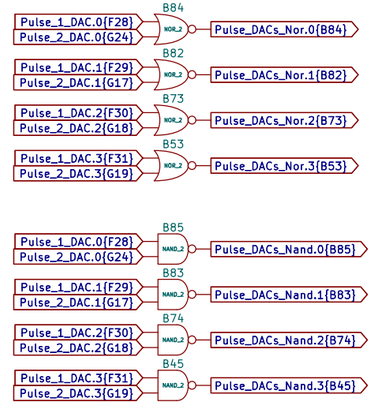


Pulse 1&2
Individual 4-bit DAC Outputs

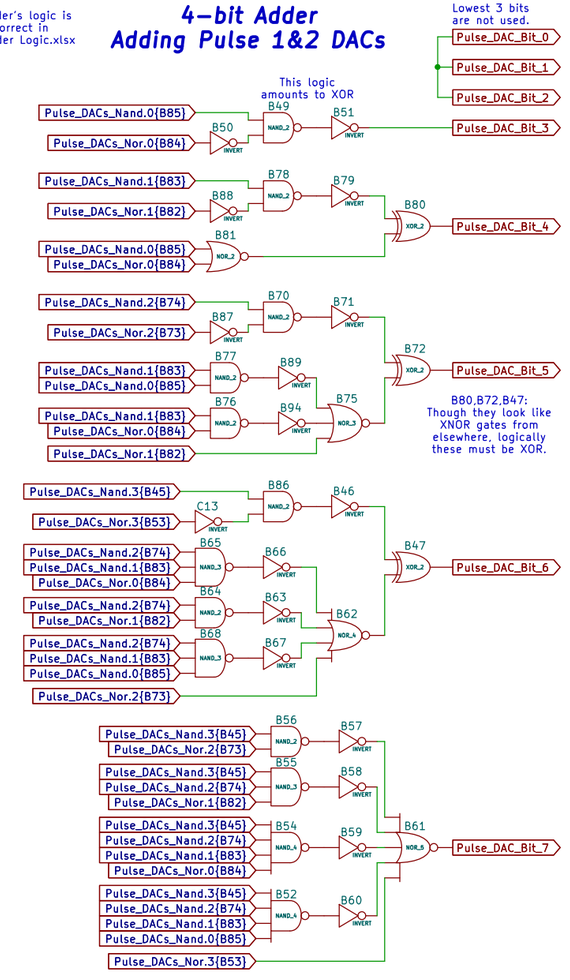


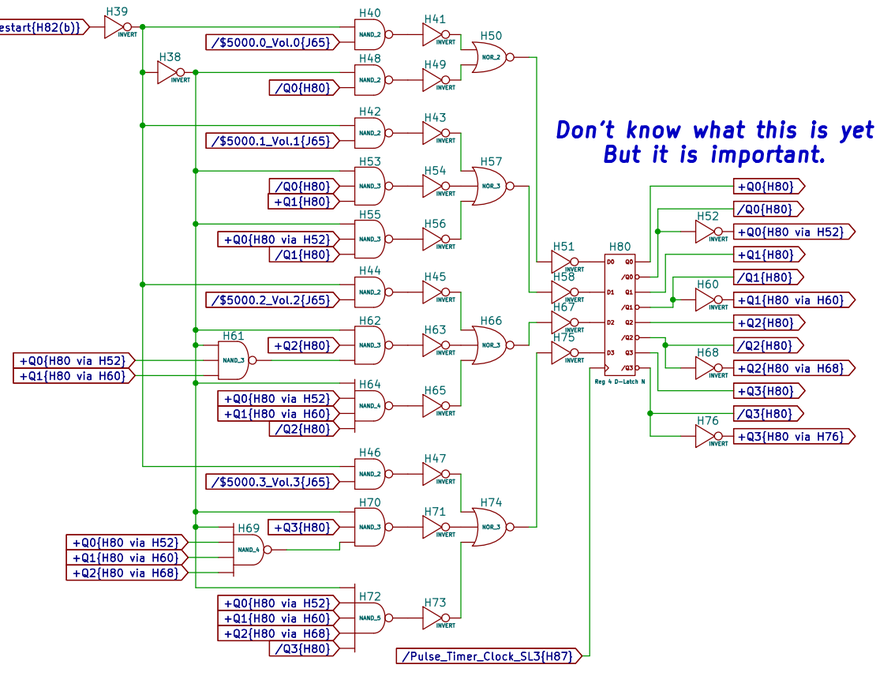
Components for
4-bit Adder

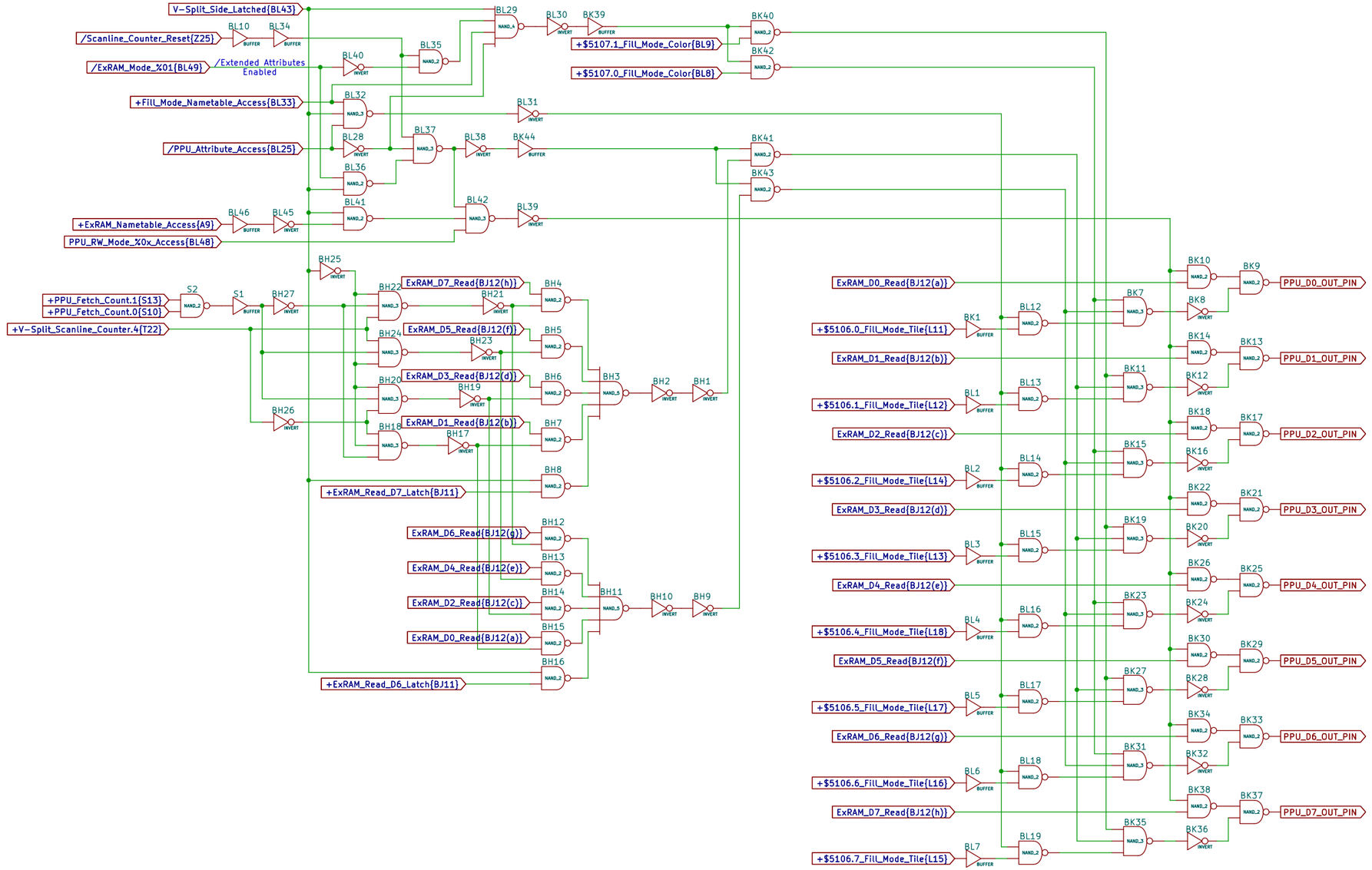
Note: The adder's logic is verified correct in Pulse DAC Adder Logic.xlsx



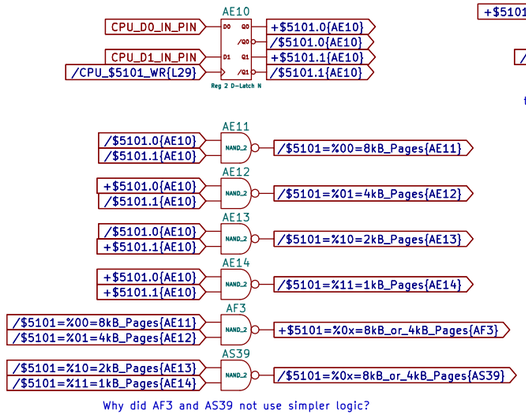
4-bit Adder
Adding Pulse 1&2 DACs



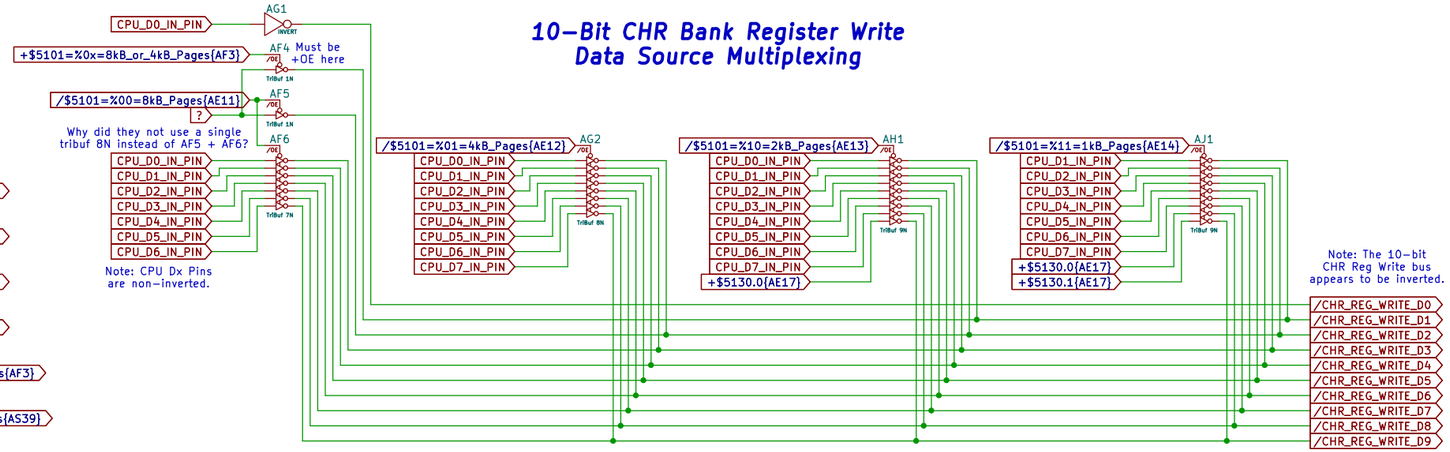



V-Split_Side_Latched{BL43}
/Scanline_Counter_Reset{Z25}
/ExRAM_Mode_%01{BL49}
/Extended Attributes Enabled
+Fill_Mode_Nametable_Access{BL33}
/PPU_Attribute_Access{BL25}
+ExRAM_Nametable_Access{A9}
PPU_RW_Mode_%0x_Access{BL48}
+PPU_Fetch_Count.1{S13}
+PPU_Fetch_Count.0{S10}
+V-Split_Scanline_Counter.4{T22}
+$5107.1_Fill_Mode_Color{BL9}
+$5107.0_Fill_Mode_Color{BL8}
ExRAM_D7_Read{BJ12(h)}
ExRAM_D5_Read{BJ12(f)}
ExRAM_D3_Read{BJ12(d)}
ExRAM_D1_Read{BJ12(b)}
+ExRAM_Read_D7_Latch{BJ11}
ExRAM_D6_Read{BJ12(g)}
ExRAM_D4_Read{BJ12(e)}
ExRAM_D2_Read{BJ12(c)}
ExRAM_D0_Read{BJ12(a)}
+ExRAM_Read_D6_Latch{BJ11}
ExRAM_D0_Read{BJ12(a)}
+$5106.0_Fill_Mode_Tile{L11}
ExRAM_D1_Read{BJ12(b)}
+$5106.1_Fill_Mode_Tile{L12}
ExRAM_D2_Read{BJ12(c)}
+$5106.2_Fill_Mode_Tile{L14}
ExRAM_D3_Read{BJ12(d)}
+$5106.3_Fill_Mode_Tile{L13}
ExRAM_D4_Read{BJ12(e)}
+$5106.4_Fill_Mode_Tile{L18}
ExRAM_D5_Read{BJ12(f)}
+$5106.5_Fill_Mode_Tile{L17}
ExRAM_D6_Read{BJ12(g)}
+$5106.6_Fill_Mode_Tile{L16}
ExRAM_D7_Read{BJ12(h)}
+$5106.7_Fill_Mode_Tile{L15}
PPU_D0_OUT_PIN
PPU_D1_OUT_PIN
PPU_D2_OUT_PIN
PPU_D3_OUT_PIN
PPU_D4_OUT_PIN
PPU_D5_OUT_PIN
PPU_D6_OUT_PIN
PPU_D7_OUT_PIN

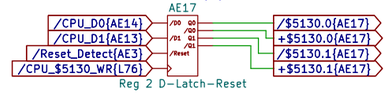
$5101 Register
CHR Mode



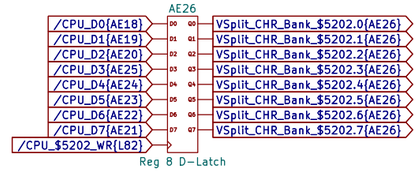
10-Bit CHR Bank Register Write
Data Source Multiplexing



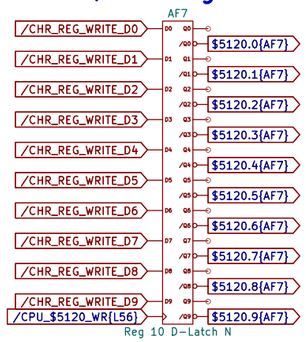
$5130 Register
CHR Bank Upper Bits



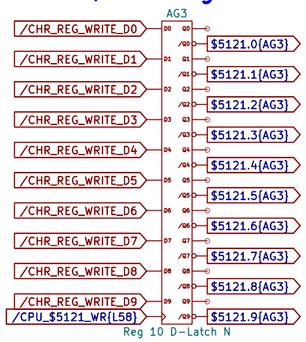
$5202 Register
V-Split CHR Bank



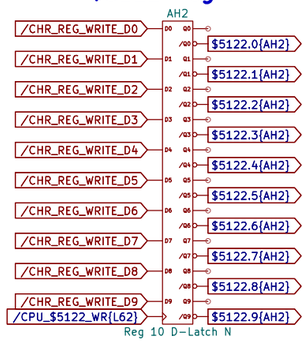
$5120 Register



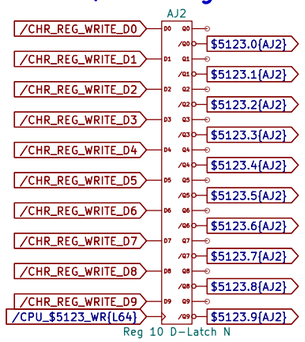
$5121 Register



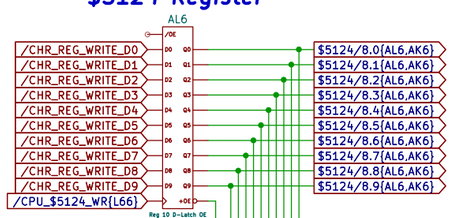
$5122 Register



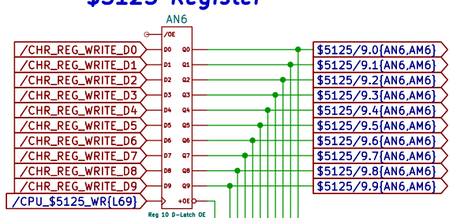
$5123 Register



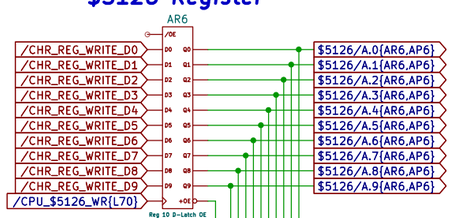
$5124 Register



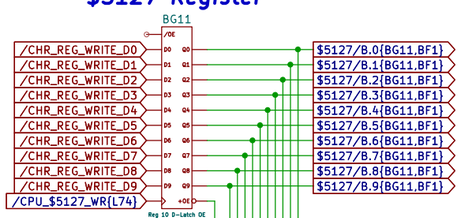
$5125 Register



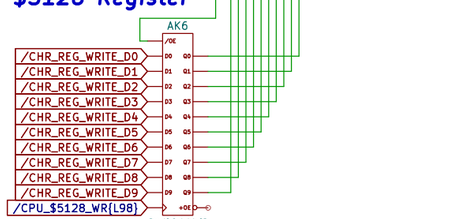
$5126 Register



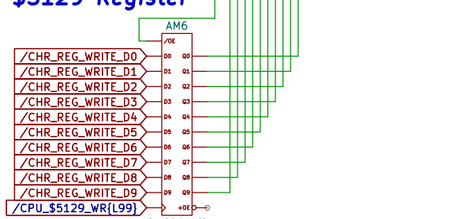
$5127 Register



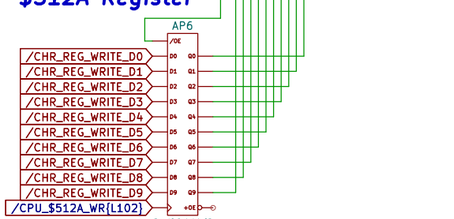
$5128 Register



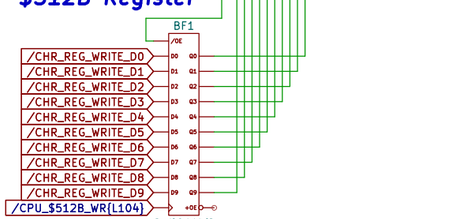
$5129 Register

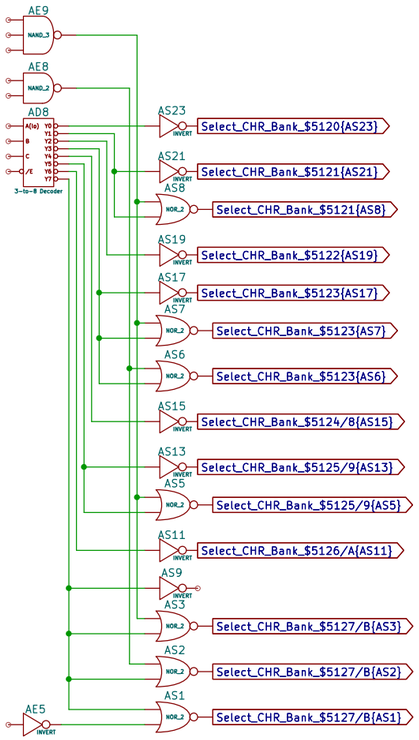


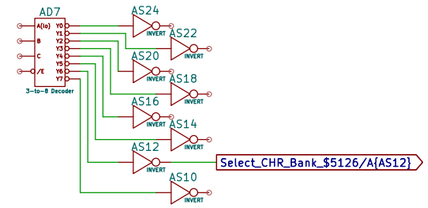
$512A Register

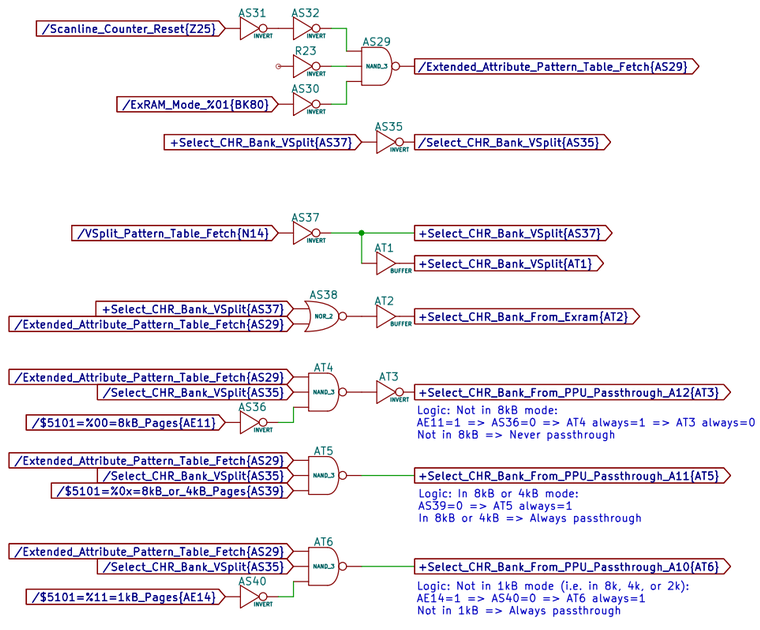


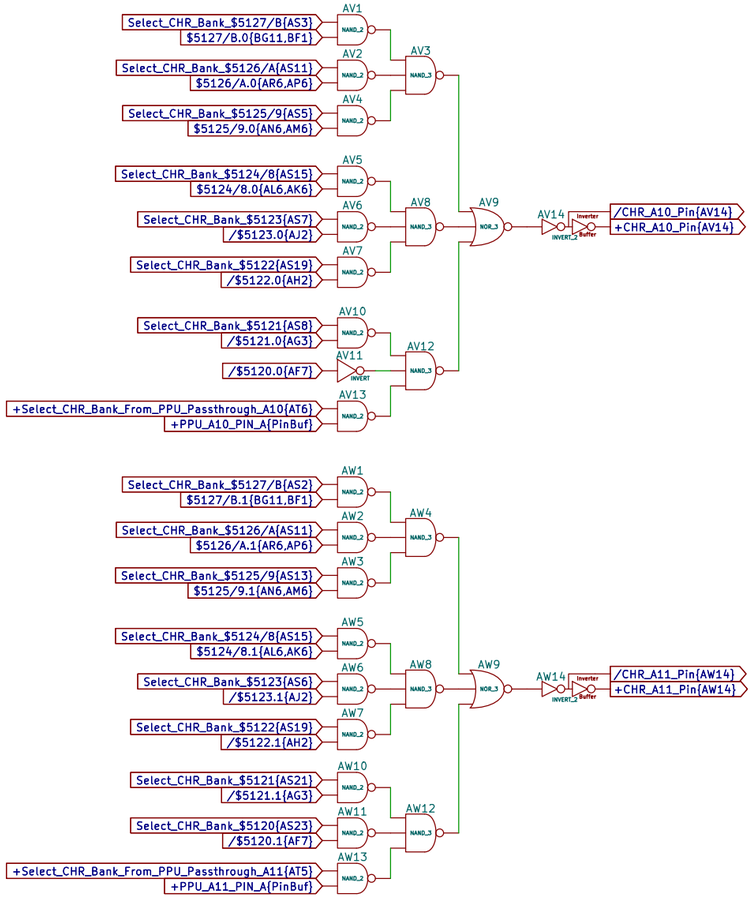
$512B Register

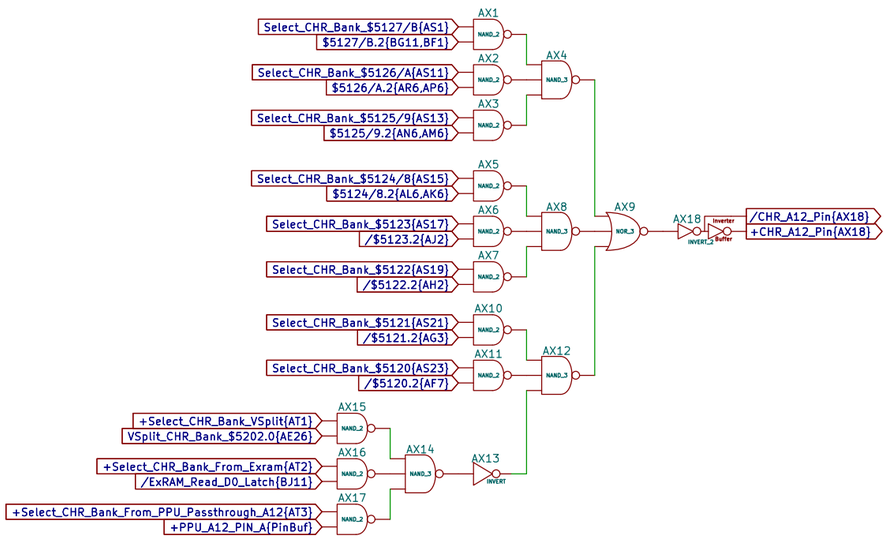



AE9
NAND_3
AE8
NAND_2
AD8
A(a)
B
C
/E
Y0
Y1
Y2
Y3
Y4
Y5
Y6
Y7
3-to-8 Decoder
AS23
INVERT
Select_CHR_Bank_$5120{AS23}
AS21
Select_CHR_Bank_$5121{AS21}
AS8
NOR_2
Select_CHR_Bank_$5121{AS8}
AS19
Select_CHR_Bank_$5122{AS19}
AS17
Select_CHR_Bank_$5123{AS17}
AS7
Select_CHR_Bank_$5123{AS7}
AS6
Select_CHR_Bank_$5123{AS6}
AS15
Select_CHR_Bank_$5124/8{AS15}
AS13
Select_CHR_Bank_$5125/9{AS13}
AS5
Select_CHR_Bank_$5125/9{AS5}
AS11
Select_CHR_Bank_$5126/A{AS11}
AS9
AS3
Select_CHR_Bank_$5127/B{AS3}
AS2
Select_CHR_Bank_$5127/B{AS2}
AS1
Select_CHR_Bank_$5127/B{AS1}
AE5


AD7
A(a)
B
C
/E
Y0
Y1
Y2
Y3
Y4
Y5
Y6
Y7
3-to-8 Decoder
AS24
INVERT
AS22
AS20
AS18
AS16
AS14
AS12
Select_CHR_Bank_$5126/A{AS12}
AS10


/Scanline_Counter_Reset{Z25}
AS31
INVERT
AS32
AS29
NAND_3
R23
AS30
/ExRAM_Mode_%01{BK80}
/Extended_Attribute_Pattern_Table_Fetch{AS29}
+Select_CHR_Bank_VSplit{AS37}
AS35
/Select_CHR_Bank_VSplit{AS35}
/VSplit_Pattern_Table_Fetch{N14}
AS37
+Select_CHR_Bank_VSplit{AS37}
AT1
BUFFER
+Select_CHR_Bank_VSplit{AT1}
AS38
NOR_2
AT2
+Select_CHR_Bank_From_Exram{AT2}
/Extended_Attribute_Pattern_Table_Fetch{AS29}
/Select_CHR_Bank_VSplit{AS35}
AT4
AT3
+Select_CHR_Bank_From_PPU_Passthrough_A12{AT3}
Logic: Not in 8kB mode:
AE11=1 => AS36=0 => AT4 always=1 => AT3 always=0
Not in 8kB => Never passthrough
/$5101=%00=8kB_Pages{AE11}
AS36
AT5
+Select_CHR_Bank_From_PPU_Passthrough_A11{AT5}
Logic: In 8kB or 4kB mode:
AS39=0 => AT5 always=1
In 8kB or 4kB => Always passthrough
/$5101=%0x=8kB_or_4kB_Pages{AS39}
AT6
+Select_CHR_Bank_From_PPU_Passthrough_A10{AT6}
Logic: Not in 1kB mode (i.e. in 8k, 4k, or 2k):
AE14=1 => AS40=0 => AT6 always=1
Not in 1kB => Always passthrough
/$5101=%11=1kB_Pages{AE14}
AS40


AV1
NAND_2
Select_CHR_Bank_$5127/B{AS3}
$5127/B.0{BG11,BF1}
AV2
NAND_2
Select_CHR_Bank_$5126/A{AS11}
$5126/A.0{AR6,AP6}
AV3
NAND_3
AV4
NAND_2
Select_CHR_Bank_$5125/9{AS5}
$5125/9.0{AN6,AM6}
AV5
NAND_2
Select_CHR_Bank_$5124/8{AS15}
$5124/8.0{AL6,AK6}
AV6
NAND_2
Select_CHR_Bank_$5123{AS7}
/$5123.0{AJ2}
AV8
NAND_3
AV9
NOR_3
AV14
INVERT
Inverter
Buffer
/CHR_A10_Pin{AV14}
+CHR_A10_Pin{AV14}
AV7
NAND_2
Select_CHR_Bank_$5122{AS19}
/$5122.0{AH2}
AV10
NAND_2
Select_CHR_Bank_$5121{AS8}
/$5121.0{AG3}
AV11
INVERT
/$5120.0{AF7}
AV12
NAND_3
AV13
NAND_2
+Select_CHR_Bank_From_PPU_Passthrough_A10{AT6}
+PPU_A10_PIN_A{PinBuf}
AW1
NAND_2
Select_CHR_Bank_$5127/B{AS2}
$5127/B.1{BG11,BF1}
AW2
NAND_2
Select_CHR_Bank_$5126/A{AS11}
$5126/A.1{AR6,AP6}
AW4
NAND_3
AW3
NAND_2
Select_CHR_Bank_$5125/9{AS13}
$5125/9.1{AN6,AM6}
AW5
NAND_2
Select_CHR_Bank_$5124/8{AS15}
$5124/8.1{AL6,AK6}
AW6
NAND_2
Select_CHR_Bank_$5123{AS6}
/$5123.1{AJ2}
AW8
NAND_3
AW9
NOR_3
AW14
INVERT
Inverter
Buffer
/CHR_A11_Pin{AW14}
+CHR_A11_Pin{AW14}
AW7
NAND_2
Select_CHR_Bank_$5122{AS19}
/$5122.1{AH2}
AW10
NAND_2
Select_CHR_Bank_$5121{AS21}
/$5121.1{AG3}
AW11
NAND_2
Select_CHR_Bank_$5120{AS23}
/$5120.1{AF7}
AW12
NAND_3
AW13
NAND_2
+Select_CHR_Bank_From_PPU_Passthrough_A11{AT5}
+PPU_A11_PIN_A{PinBuf}


AX1
NAND_2
Select_CHR_Bank_$5127/B{AS1}
$5127/B.2{BG11,BF1}
AX2
NAND_2
Select_CHR_Bank_$5126/A{AS11}
$5126/A.2{AR6,AP6}
AX4
NAND_3
AX3
NAND_2
Select_CHR_Bank_$5125/9{AS13}
$5125/9.2{AN6,AM6}
AX5
NAND_2
Select_CHR_Bank_$5124/8{AS15}
$5124/8.2{AL6,AK6}
AX6
NAND_2
Select_CHR_Bank_$5123{AS17}
/$5123.2{AJ2}
AX8
NAND_3
AX9
NOR_3
AX18
INVERT
Inverter
Buffer
/CHR_A12_Pin{AX18}
+CHR_A12_Pin{AX18}
AX7
NAND_2
Select_CHR_Bank_$5122{AS19}
/$5122.2{AH2}
AX10
NAND_2
Select_CHR_Bank_$5121{AS21}
/$5121.2{AG3}
AX11
NAND_2
Select_CHR_Bank_$5120{AS23}
/$5120.2{AF7}
AX12
NAND_3
AX15
NAND_2
+Select_CHR_Bank_VSplit{AT1}
VSplit_CHR_Bank_$5202.0{AE26}
AX16
NAND_2
+Select_CHR_Bank_From_Exram{AT2}
/ExRAM_Read_D0_Latch{BJ11}
AX14
NAND_3
AX13
INVERT
AX17
NAND_2
+Select_CHR_Bank_From_PPU_Passthrough_A12{AT3}
+PPU_A12_PIN_A{PinBuf}